

Day 4 - Intro to the CLI

Thursday, August 20, 2026

3:28 PM

Mode	Prompt	Purpose	How to Enter
User EXEC	Router>	Basic monitoring/access	Default
Privileged EXEC	Router#	Advanced monitoring and device management	enable
Global Configuration	Router(config)#	Configure the device	configure terminal

Term / Command	What It Does	Example
?	Displays available commands/options	Router# ?
command ?	Shows available options for a specific command	Router# show ?
TAB	Automatically completes a partially typed command	Router# conf → TAB → configure
enable	Moves from User EXEC → Privileged EXEC	Router> enable
disable	Moves from Privileged EXEC → User EXEC	Router# disable
configure terminal	Enters Global Configuration Mode	Router# configure terminal
conf t	Short version of configure terminal	Router# conf t
exit	Moves back one CLI level	Router(config)# exit → Router#
end	Immediately returns to Privileged EXEC Mode	Router(config-if)# end → Router#
do	Allows you to run a Privileged EXEC command while in configuration mode	Router(config)# do show running-config
enable password	Sets a password for entering Privileged EXEC mode	Router(config)# enable password cisco
enable secret	Sets an encrypted password for Privileged EXEC mode; preferred over enable password (uses MD5 More secure encryption)	Router(config)# enable secret MyPassword
service password-encryption	Encrypts plaintext passwords in the configuration using Cisco Type 7 encryption (weak encryption)	Router(config)# service password-encryption
no service password-encryption	Stops future plaintext passwords from being automatically encrypted; existing Type 7 passwords remain encrypted	Router(config)# no service password-encryption
show running-config	Displays the current active configuration	Router# show running-config
show startup-config	Displays the configuration saved in NVRAM that will be loaded after reboot	Router# show startup-config
write	Saves the running configuration to startup configuration	Router# write
write memory	Saves the running configuration to startup configuration	Router# write memory
copy running-config startup-config	Copies the active configuration into the startup configuration	Router# copy running-config startup-config
no	Removes/disables a configuration command	Router(config)# no service password-encryption

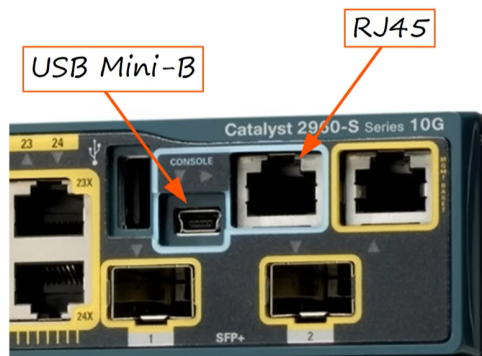
What is a CLI?

- A "Command-line Interface"
- The interface you use to configure Cisco devices

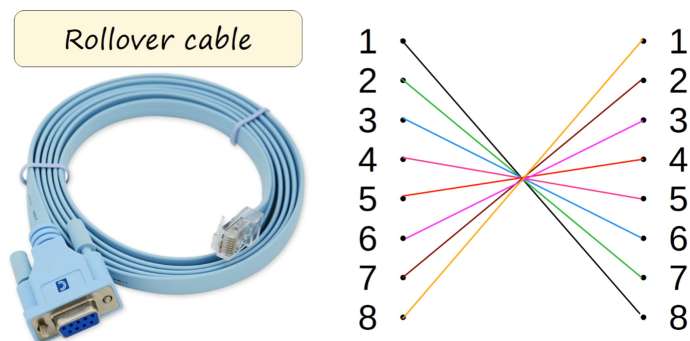
A GUI is a "Graphical User Interface"

How do you connect to a Cisco Device?

- Console Port : When you first configure a device, you have to connect via the Console Port. You can use a "Rollover cable" : DB9 serial connector to RJ45 OR a DB9 Serial to USB



How to connect to a Cisco device? (Console port)



How do you actually access the CLI?

- You need to use a TERMINAL EMULATOR (Example: PuTTY is a popular choice) and connect via "Serial" (default settings)

Cisco Default Settings are:

Speed (baud) : 9600 bits/second

Data bits: 8 data bits

Stop bits: 1 stop bit (sent after 8 data bits are sent)

Parity: None

Flow Control: None

CLI:

When you first enter the CLI you will DEFAULT be in what is called 'User EXEC' mode.

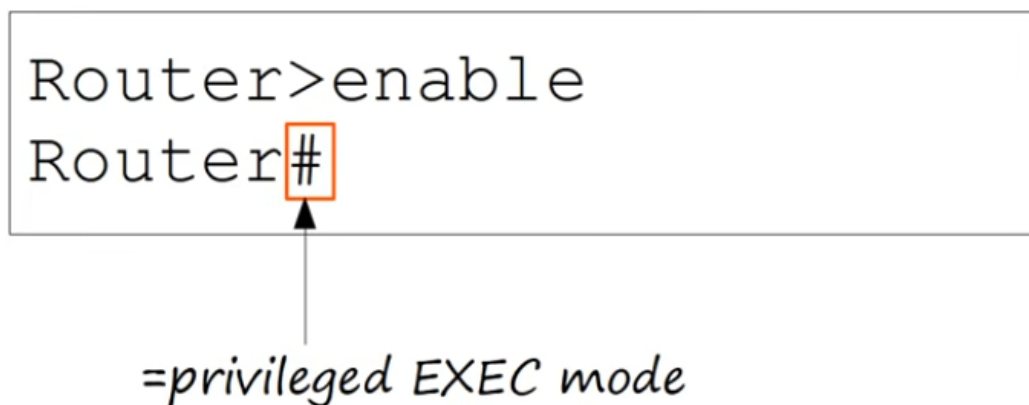
USER EXEC MODE:





- User EXEC mode is very limited.
- User can look at some things but can't make ANY changes to the configuration.
- AKA 'User Mode'

Privileged EXEC



- Provides complete access to view the device's configuration, restart the device, etc.
- Cannot change the configuration, but can change the time on the device, save the configuration file, etc.

Help

USE a Question Mark (?) to view the available commands in ANY mode.

Combining ? with a letter or partial command will list all the commands with those letters.

Using a ? After a command will give you available options for using that command.

Tab

USE the TAB key to complete partially entered commands IF the command exists.

Example entering **en (short for enable)** will enter you to privilege exec mode.

Do

Do allows you to run privilege exec commands in other levels

No

Used to remove a command you previously used

GLOBAL CONFIGURATION MODE:

To enter Global Configuration Mode, enter the command, within Privileged EXEC mode
'configure terminal' (or 'conf t')

Router# configure terminal

Type 'exit' to drop back into 'Privileged EXEC' mode.

To Enable Password for User EXEC mode:

Router(config)# enable password (password)

- Passwords ARE case-sensitive.

// This command encrypts plain-text passwords, visible in the config files, using simple encryption.

Router(config)# service password-encryption

If you enable 'service password-encryption'

- Current passwords WILL be encrypted.
- Future passwords WILL be encrypted.
- The 'enable secret' WILL NOT be effected.

If you disable 'service password-encryption'

- Current passwords WILL NOT be decrypted.
- Future passwords WILL NOT be encrypted.
- The 'enable secret' WILL NOT be effected.

// This command enables passwords for the Privileged EXEC mode.

Router(config)# enable secret (password)

// enable secret will ALWAYS be encrypted (at level 5)

There are TWO separate configuration files kept on the device at once.

Running-config :

- The current, ACTIVE configuration file on the device. As you enter commands in the CLI, you edit the active configuration.

Startup-config :

- The configuration file that will be loaded upon RESTART of the device.

To see the configuration files, inside 'Privileged EXEC' mode:

Router# show running-config // for running config //

OR

Router# show startup-config // for startup config //

To SAVE the Running configuration file, you can:

Router# write Building configuration... [OK]

Router# write memory Building configuration... [OK]

Router# copy running-config startup-config

Destination filename [startup-config]?

Building configuration... [OK]

To encrypt passwords:

Router# conf t

Router(config)# service password-encryption

Router(config)# service password-encryption

This makes all current passwords *encrypted*

Future passwords will ALSO be *encrypted*

“Enable secret” will not be effected (it’s ALWAYS encrypted)

```
Router#show running-config
Building configuration...

Current configuration : 719 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable password 7 08026F6028
!
```

Now you will see that the password is no longer in plaintext.

“7” refers to the type of encryption used to encrypt the password. In this case, “7” uses Cisco’s proprietary encryption.

“7” is fairly easy to crack since the encryption is weak.

For BETTER / STRONGER encryption, use “enable secret”

```
Router(config)#enable secret Cisco
Router(config)#do sh run
Building configuration...

Current configuration : 766 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$YlCkLMcTYWwkF1Ccndt1l.
enable password 7 08026F6028
```

“5” refers to MD5 encryption.

Can still be cracked but it’s much much stronger.

Once you use “enable secret” command, this will override “enable password”

To CANCEL or delete a command you entered, use the “no” keyword

```
Router(config)#no service password-encryption
Router(config)#do sh run
```

```

Building configuration...

Current configuration : 769 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$YlCkLMcTYWwkF1Ccndt1l.
enable password 7 08026F6028
!

```

In this instance, disabling “service password-encryption”:

- current passwords will NOT be decrypted (unchanged)
- future passwords will NOT be encrypted
- the “enable secret” will not be effected



Modes Review

Router> *= user EXEC mode*

Router# *= privileged EXEC mode*

Router(config)# *= global configuration mode*



Command Review

Router>**enable**
##used to enter privileged EXEC mode

Router#**configure terminal**
##used to enter global configuration mode

##used to enter global configuration mode

Router(config)#**enable password** password

##configures a password to protect privileged exec mode



Command Review

Router(config)#**service password-encryption**

##encrypts the enable password (and other passwords)

Router(config)#**enable secret** password

##configures a more secure, always-encrypted enable password

Router(config)#**run** privileged-exec-level-command

##executes a privileged-exec level command from global configuration mode



Command Review

Router(config)#**no** command

##removes the command

Router(config)#**show running-config**

##displays the current, active configuration file

Router(config)#**show startup-config**

##displays the saved configuration file which will be loaded if the device is restarted



Command Review

```
Router(config)#write  
##saves the configuration
```

```
Router(config)#write memory  
##saves the configuration
```

```
Router(config)#copy running-config startup-config  
##saves the configuration
```